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Quantum Computing in US Banking: The Future of Fraud Prevention and Financial Crime Detection

Ashok Ghimire[1*]

[1]Westcliff University, USA

[1]ashok.ghimire1991@gmail.com

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Corresponding Author

Ashok Ghimire

ashok.ghimire1991@gmail.com

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ABSTRACT

The integration of Artificial Intelligence (AI) in food production is revolutionizing the industry by enhancing efficiency, improving food safety, and driving sustainability. Smart food factories powered by AI are optimizing production processes through automation, predictive maintenance, and real-time quality control. AI-driven supply chain management is reducing food waste, ensuring better resource allocation, and streamlining logistics. Furthermore, AI is playing a crucial role in developing personalized nutrition and alternative protein sources, catering to evolving consumer demands. Despite its numerous benefits, AI adoption in food manufacturing faces challenges such as high implementation costs, data privacy concerns, and workforce displacement. Overcoming these obstacles requires investment in AI training, regulatory frameworks, and ethical AI deployment. Looking ahead, advancements in robotics, block chain integration, and AI-powered 3D food printing will further shape the future of food production. By addressing these challenges and leveraging AI responsibly, the food industry can create safer, more efficient, and sustainable food production systems for the future.

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INTRODUCTION

Criminals have traditionally been able to exploit vulnerabilities on the part of the financial sector and banks. In the United States, banks handle billions of transactions daily, making them attractive to fraudsters, cybercriminals, and money launderers. Since financial institutions are increasingly embracing digital technologies, financial crime risk has grown and turned into cyber threats that are more complex and require more advanced security measures. However, financial crime in US banking has a broad range of definition which includes identity theft, credit card fraud, wire fraud, money laundering and insider trading [1]. It's getting increasingly difficult to detect fraud in the context of digital banking, mobile payments and crypto currency with criminals using sophisticated techniques like deep fake identity fraud, phishing attacks and ransom ware to bypass even established security measures. In turn, banks have sunk money into fraud detection systems, artificial intelligence (AI) and block chain technology to beef up defenses against such attacks. But, conventional computing has limitations, especially when it comes to interactively querying and processing massive data at high-speed [2].

Financial crime is becoming more sophisticated, and existing security frameworks cannot provide an answer to this problem. Quantum computing is where this all comes into play. Whereas traditional computers work using binary all bits (0s and 1s), quantum computers operate on quantum all bits (0s and 1s) that allows them to perform calculations at an exponentially faster rate [3]. The speed and computational power have the power to transform banking by serving as a tool for fraud detection, real time risk analysis, and even encryption techniques in the banking industry. Although promise, a quantitative integration of quantum computing into the financial sector is not without thorny challenges. However, banking adoption can only come with time, massive investment, regulatory adjustment, along with a strategic approach to overcome the risks involved, as the technology is still in its early stages [4]. Moreover, with the advancement of quantum computing, it can also compromise existing cryptographic security measures, rendering current encryption methods as obsolete as the information can be exposed to cyber-attacks powered by quantum [5].

The problem facing US banks is at a crucial moment of financial security. On the one hand, they must get ready to face the challenges that quantum computing can bring about, while on the other hand, they have a rare chance to utilize its strengths in aiding fraud prevention. Services like developing

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quantum resistant encryption, implementing advanced fraud detection models, or working with technology firms to lead ahead of financial criminals are included in this [6]. With quantum computing becoming more sophisticated, US banks need to begin adapting to stay resilient in a competitive and ever more complicated high stakes cybersecurity environment.

QUANTUM COMPUTING: UNDERSTANDING A GAME CHANGER FOR FRAUD PREVENTION

An emerging technology of the 21st century, quantum computing, is already poised to revolutionize banks and financial safety. Whereas conventional computers process and store data in binary form (1s and 0s), quantum computers utilize quantum bits (qubits) which can be in a state of multiple states simultaneously thanks to the phenomena of superposition and entaglement. As a result, quantum computers can process thousands of such bits, more than any standard computer can do, and solve problems that consuming millions of years would require using ordinary computing methods [7].

The opportunities for US banks associated with preventing Fraud Detection and ensuring Financial Security related to the real time analysis of enormous data sets through quantum computing cannot be underestimated. Often, financial fraud involves identifying subtle patterns in enormous transaction records – something that blurs to even the most advanced classical computing systems. As for banks, they can use quantum computing to analyze and process massive data sets at such speeds that fraud detection accuracy increased and false positives reduced [8].

Making use of quantum computing, especially in anomaly detection to prevent fraud appears to be one of the most promising applications. Historically, traditional fraud detection has consumed rule based algorithms and machine learning model which compare transactions against historic data to flag suspicious activity. But the sophisticated fraudsters change their tactics constantly and trick a lot of conventional models along the way [9]. The power of quantum machine learning (QML) algorithms lie in that they can be run on a quantum computer for faster and accurate detection of hidden correlations, including the identification of fraudulent transactions that Classical systems might not catch. QML would also enable US banks to analyses transaction patterns in real time so they could identify fraudulent activity before it happens [10].

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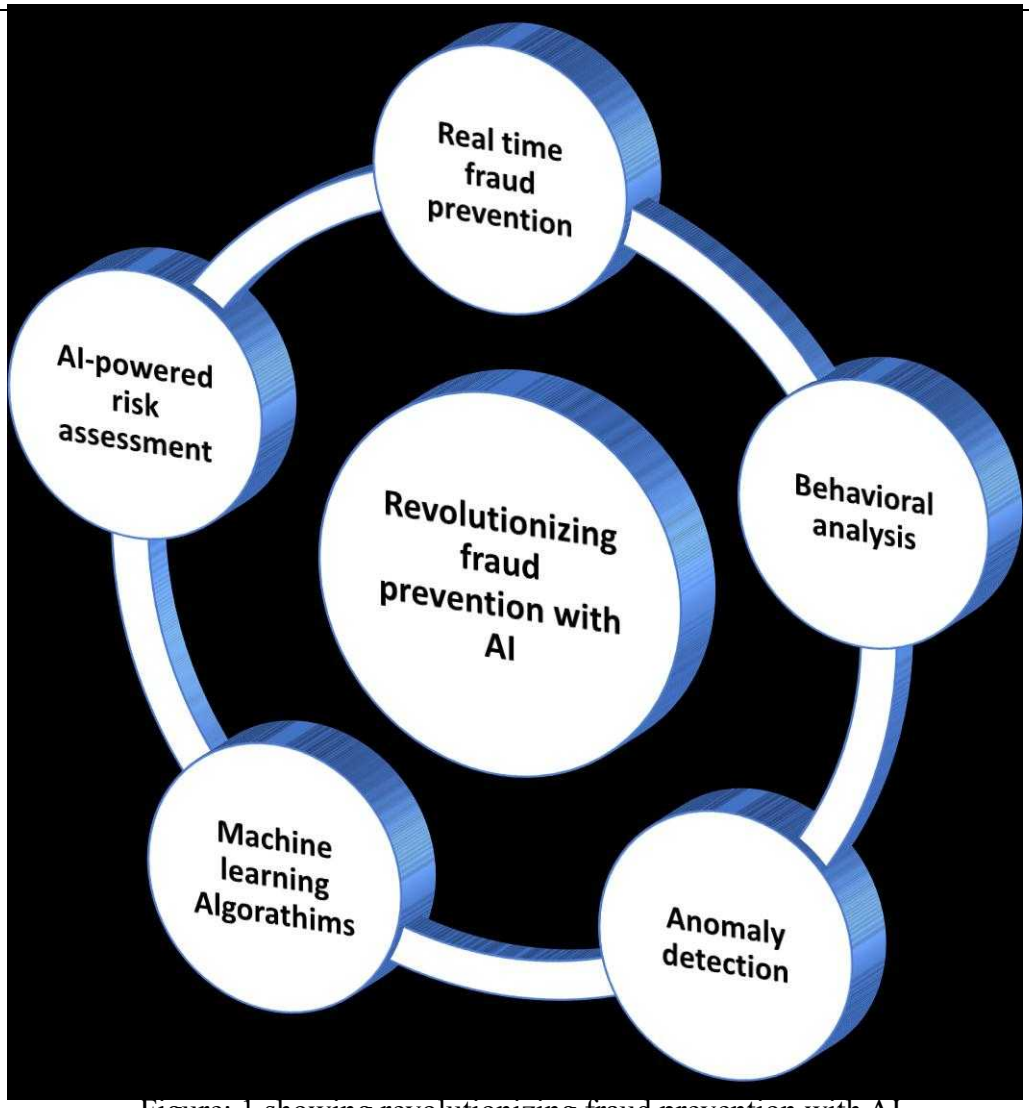

Figure: 1 showing revolutionizing fraud prevention with AI

In another significant advantage, the use of quantum computing can improve cryptographic security. Encryption is heavily relied on in financial transactions as it protects sensitive data in the transactions but current encryption methods like RSA and ECC (Elliptic Curve Cryptography) can be made vulnerable once large scale quantum computers are developed. This is because an encryption algorithm such as that used by modern electronics can be broken by Shor's algorithm on a quantum computer (but not a classical one), because it factors large prime numbers exponentially faster than classical computers [11]. In order to prevent this threat, US banks have begun to research quantum resilient encryption methods, including lattice based cryptography and hash based signatures as a means of encryption for their financial data against future quantum powered cyber-attacks [12]. At the same time, quantum computing plays a supportive role in risk management and predictive

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modeling. Credit risk, market movements, default risks are all modeled by banks through complex models. Banks can use quantum algorithms to optimize these models because their quants allows for studying multiple variables at once, estimating more accurate risk and assisting banks in making better informed decisions [13].

Although there is direct potential for quantum computing, no one has truly implemented its full capability in financial fraud prevention yet. The barriers to its adoption are: a high cost of development, the need for specialized quantum hardware, and regulatory issues. But as quantum technology continues to improve and it becomes a more accessible technology, US banks must begin to prepare for when it inevitably affects financial security [14]. Investing in quantum research, working with tech companies and deploying quantum resistant security would equip banks to stay ahead of the curve, and ultimately, a safer financial future.

CURRENT CHALLENGES IN FRAUD DETECTION AND FINANCIAL CRIME PREVENTION

The challenges faced by the US banking in the area of financial crime are developing constantly and fraudsters do not stop inventing new and more sophisticated techniques to bend the banking systems. As digital banking, real time payments and adopting crypto currency is increasing financial institutions are fighting against growing number of threats, from simple identity theft to highly complex coordinated cyber-attacks. Although fraud detection has advanced, there is still a lot of work to be done in order to keep ahead of new threats. In banking, the number of fraudulent activities has gone beyond the traditional credit card fraud or check forgery. Today's criminals use highly advanced technologies like artificial intelligence (AI), deep fake identities, and bot networks to get through the old patterns of security [15]. Banks and customers are vulnerable to phishing, social engineering and synthetic identity fraud, rendering fraud detection that much more complex for the security teams. In addition, the number of ransom ware attacks against financial institutions has increased, where hackers demand a ransom, to restore access to critical systems [16].

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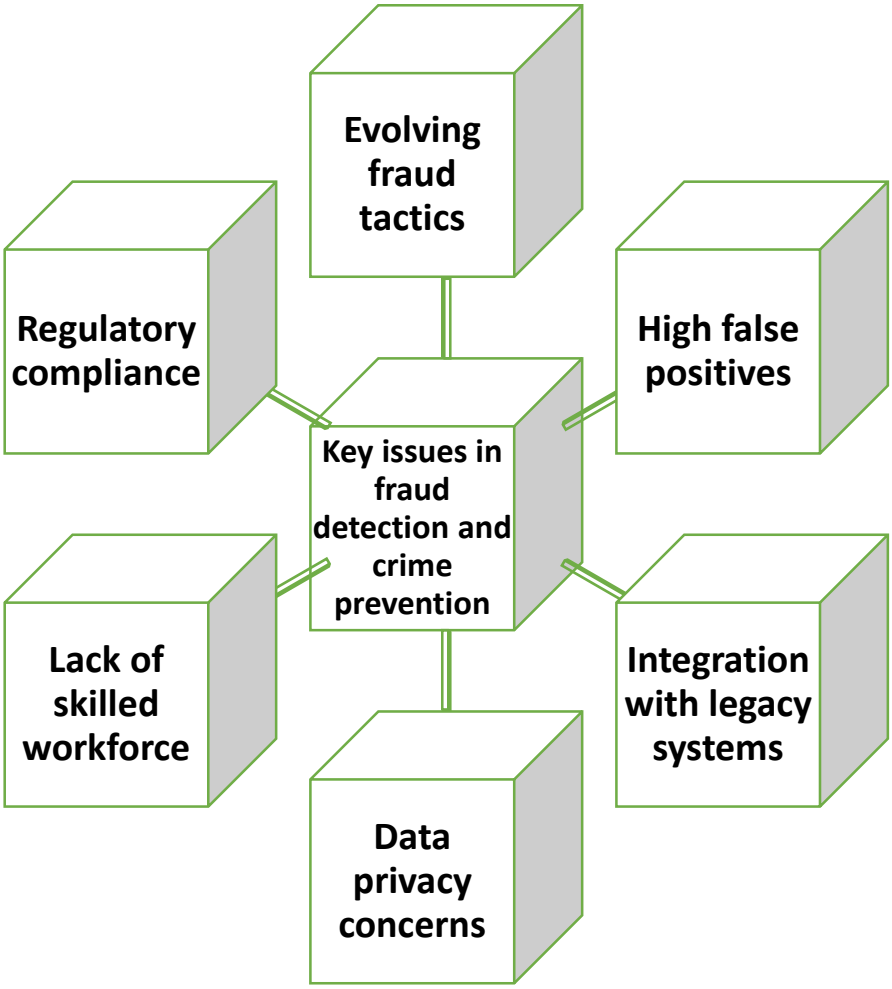

Figure: 2 showing key issues in fraud detection

LIMITATIONS OF TRADITIONAL FRAUD DETECTION SYSTEMS

Generally, most banks use rule based systems and machine learning algorithms to catch the fraud by flagging the suspicious transactions on the basis of patterns that are pre-defined. But these systems have typically had trouble with: Many legitimate transactions become marked as fraudulent, which results in customer discontent and operational inefficiencies [17].

False negatives: There instances whereby some fraudulent activities are not foreseen because fraudsters are continually coming up with new ways of mimicking normal behavior [18].

Limited Response Time: Fraud detection models in use today could take days or weeks to process

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transaction volumes large enough for their statistical weight to really matter. They also rely on traditional security infrastructures through RSA and ECC (Elliptic Curve Cryptography), which will be certainly ineffective when quantum computers will be powerful enough to break them. The future risk for data security in financial institutions is very high [19].

The use of real time payment systems like the Federal Reserve's FedNow service enables customers to execute money transfers in real time. This is convenient. Unfortunately, this also enables fraudsters to conduct exploitative but swift attacks to carry out fraudulent transactions while banks are looking for them to block [20]. Unlike traditional banking, where you are able to tag suspicious transfers and reverse their transaction within a matter of hours, real time payment, removes any kind of ability to reverse a transaction after funds have been sent.

Fraud does not always have to be orchestrated by an external hacker as it is possible to trace it to the insider threats: US banks are equally vulnerable. Sensitive financial data that employees have access to may be exploited by employees to gain something personally, or compromised data may serve as a source of information for fraud rings outside the organization. Weak internal controls, poor monitoring, and failure to perform the proper background checks on employees are leading to this ongoing problem [21].

Regulatory and Compliance Challenges: The US banks are subject to stringent regulatory framework aimed towards prevention of fraud such as Bank Secrecy Act (BSA), Anti Money Laundering (AML), Know Your Customer (KYC), etc. However, the need to comply with these regulations also involves resource investment and in most cases, continues to be a step behind financial crime tactics [22]. Security measures must be tight at banks, but it cannot result in customer experiences that are unnecessarily delayed, in onboarding unnecessarily, in executing transactions unnecessarily.

The Emergence of Some Opportunities and Threats: Quantities computing opens up opportunities for fraud prevention, but also introduces new threats. Sensitive banking data can be put at risk if quantum computers get strong enough to crack existing encryption standards. Legal entities in the US, such as banks, need to adopt post quantum cryptography in order to prevent them from becoming vulnerable to quantum powered cyber-attacks in the future [23]. Despite fraud detection and

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cybersecurity advancements, US banks still find it difficult to stop financial crime. As complexity of fraud tactics and sophistication of fraudsters keep increasing and sophistication of fraud detection approaches continues to create limitations, and as real time transactions are growing at an unrestrained rate, it is impossible to keep up with the constantly changing fraud threats. Meanwhile, the land is already littered with the risks of insider fraud, regulatory burdens, and appalling creaks of growing threats from quantum computing [24]. Financial institutions need to invest in the next generation technologies, like AI based fraud detection, block chain security and quantum resistant encryption to stay in the game. US banks can be on the front foot of these challenges in order to improve fraud prevention and protect more customers in a rapidly digitizing higher risk financial environment [25].

Ever escalating sophistication of financial fraud forces US banks to switch to the edge technologies to enhance security and protect their customers. The revolutionary computational power of quantum computing can be used to prevent fraud by detecting fraudulent activities in real time, improving encryption methods, and optimizing the risk management. Banks can use quantum capabilities to carry out complex processing of massive amounts of transactions with greater accuracy and can uncover complex fraud patterns that are very hard compute using traditional computing [26]. Encryption is one of the most important aspects of financial security, it should protect sensitive data of the customer, details of transactions as well as financial systems from all possible threats in cyberspace. But, encryption techniques like RSA (RivestShamirAdleman) and ECC (Elliptic Curve Cryptography) become vulnerable to being broken by quantum computers using Shor's Algorithm — a quantum algorithm which can factor large prime numbers much faster than classical computers, exponentially [27].

OPPORTUNITIES FOR US BANKS: THE FUTURE OF FINANCIAL SECURITY

With financial frauds becoming increasingly sophisticated, US banks have an opportunity to make use of quantum computing for better security and fraud prevention. Despite the difficulties of implementing quantum technology, the risks of the technology leave much to be desired in comparison to the actual benefits. Advancements in cybersecurity, predictive fraud detection, and even collaborations with tech firms) can be possible through quantum computing, ensuring that financial systems and other sectors involved are future proof [28].

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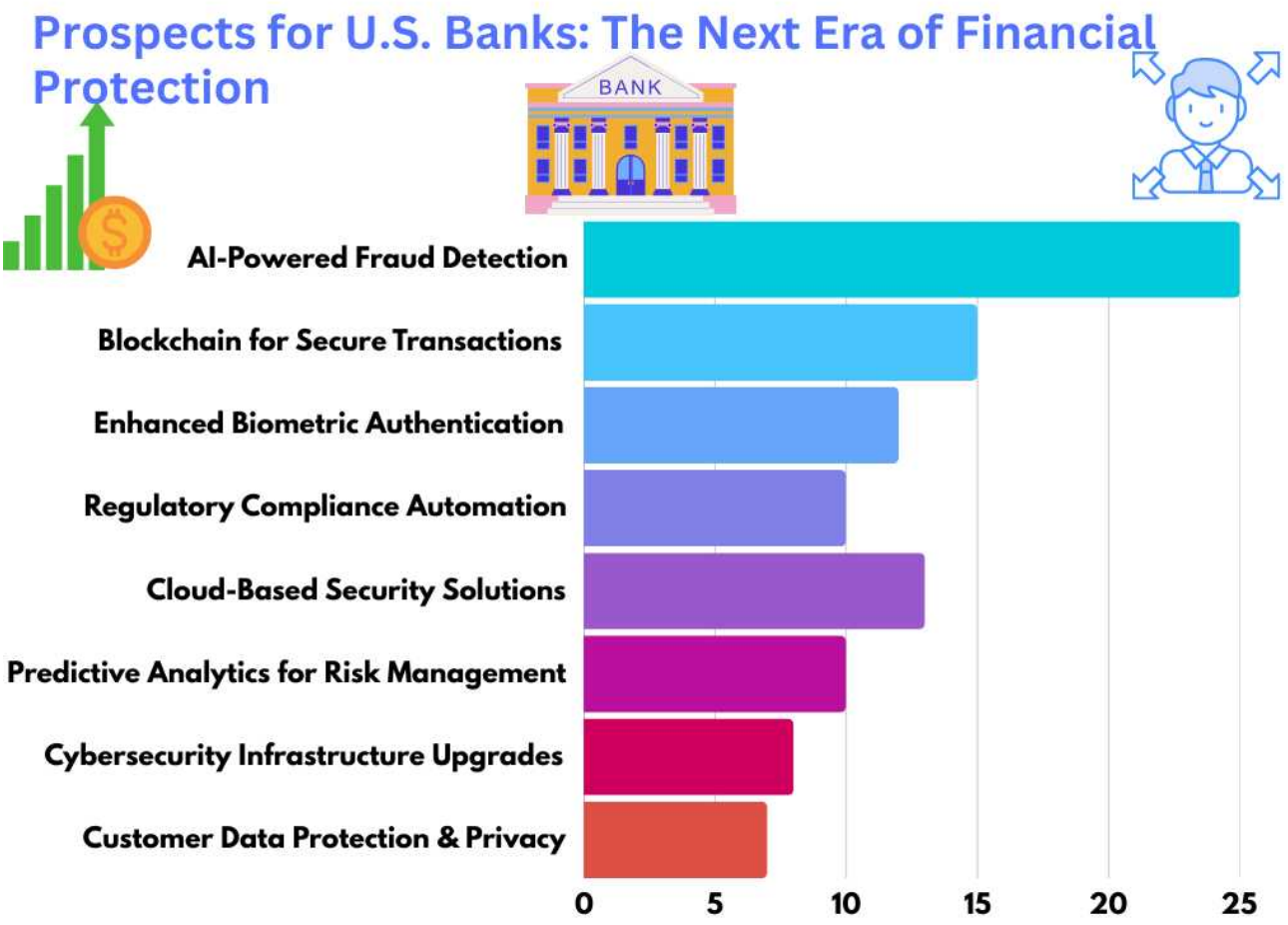

Figure: 3 showing prospects of US banks the next era financial protection

The main benefit of quantum computing is that it can boost security to defend cybersecurity. RSA and ECC being traditional encryption methods are susceptible to the quantum decryption. To combat this, banks have to move to quantum resistant cryptographic methods, including: Lattice-based cryptography Resistant to quantum attacks and already under evaluation for widespread adoption. Creates encryption keys using the mysteries of quantum mechanics, and any attempt by hackers to eavesdrop is immediately detectable [29].

New algorithms that are designed to work even in a post quantum world, for the safety of banking transactions. US banks should integrate these solutions within their cybersecurity frameworks to lead quantum enabled cyber-attack and stop large scale data breaches. Currently, AI and machine learning models are used for fraud detection today, but because of the limitations of current computing power,

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it is not possible for them to accurately crunch the big data in real time [30]. This is changed by quantum computing which allows:

Quantum algorithms process massive datasets almost instantaneously and can recognize a fraudulent attempt that traditional systems would miss. Quantum powered AI can model many fraud scenarios at the same time and thus improve the risk predictions in banking transactions. Quantum AI is adaptive fraud detection – ensuring that fraud detection measures always one step ahead of fraudsters. If implemented, US banks can use quantum AI to eliminate mistakes in fraud detection and reduce the number of false positives and negatives by leaps and bounds [31].

COLLABORATION BETWEEN BANKS AND TECH INNOVATORS

Practicing quantum computing is not for the faint of heart as it demands expertise and big infrastructure and the banks, therefore, have to work with tech companies, cybersecurity firms, and government agencies in order to be able to accelerate adoption [32]. Strategic partnerships can offer:

Cutting edge quantum research available to unique quantum computing leaders, such as IBM, Google, as well as startups that focus on financial security solutions. A joint development of quantum fraud prevention tools – Quantum powered fraud detection models can be used by banks in conjunction with its preexisting security systems. Regulatory bodies and banking consortia must lay out a clear framework for (widespread) adoption of quantum technology for the security of electronic transactions [33].

Quantum computing is set to mature, and US banks that get involved early in quantum collaborations will have the competitive advantage in fraud prevention and financial security. US banks can leverage quantum computing to provide transformational opportunities in the form of improved cybersecurity, predictive fraud prevention, and new industry collaboration using quantum computing. Financial institutions that are readying themselves for the adoption of quantum now can future proof their security infrastructure for more secure and more effective fraud prevention for years to come [34].

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CHALLENGES AND RISKS OF QUANTUM ADOPTION IN US BANKING

Quantum computing is an incredible opportunity for organizations involved in improving financial security and reducing fraud, and will have massive potential in US banking, but this is not going to be without huge challenges and risks in the US banking channel. Financial institutions hoping to incorporate quantum technology must wade through issues from high implementation costs to new cybersecurity threats to regulatory hurdles [35]. Implementing quantum computing is one of the biggest challenges for US banks due to its cost and complexity. Quantum computers are not similar to conventional computing systems that require specialized hardware, extreme cooling systems and highly trained professionals for development and maintenance [36]. The barriers include:

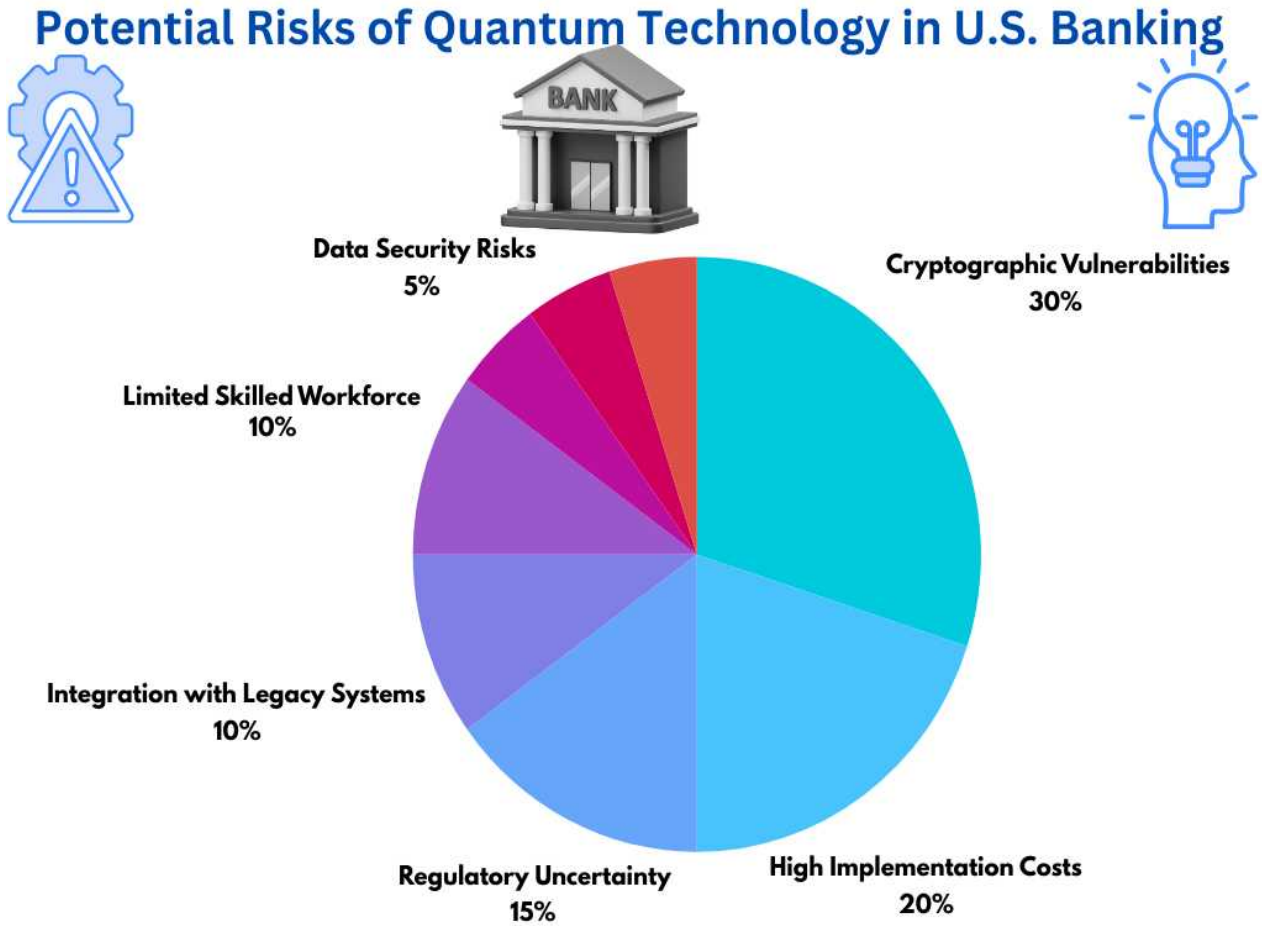

Figure: 4 showing potential risks quantum technology in US banking

Quantum computers are expensive to set up, requiring set up of expensive infrastructure – cryogenic temperatures and highly controlled environments. Currently, only a few tech companies, such as IBM,

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Google and D-Wave, provide limited access to quantum computing via cloud based services. This may be less than suitable for sensitive financial data. Quantum computing expertise is scarce, therefore banks will have to spend massively both in hiring and training talent. For many, especially smaller banks, could become a major hurdle, as it is potentially too large of an upfront investment for many financial institutions [37].

Quantum-Resistant Threats from Hackers: Quantum computing not only boosts fraud detection, but also brings new risks for security. The biggest issue is that quantum machines could use their power to perpetrate cyberattacks. Therefore, hackers with access to quantum computers could break traditional encryption systems and therefore expose sensitive banking data. In breaking RSA and ECC encryption, that means the cryptographic methods on which current banking security is based could become obsolete if large-scale quantum computers are ever created [38]. This will include quantum technology being used by cybercriminals to outpace fraud detection systems at speed unheard of before.

Attackers will have access at quantum scale, however, the actual volume and speed of exploitation from the financial networks is far more than today's threat. To protect against these, banks will need to start preparing for a shift towards post quantum cryptography (PQC) that is standards of encryption that our quantum computers cannot attack. PQC adoption is still in infancy across the financial sector; however [39].

REGULATORY AND COMPLIANCE CONSIDERATIONS

Regulatory and legal challenges pose to quantum computing in banking. There are several strict security and privacy regulations that financial institutions must comply with. The Bank Secrecy Act (BSA) which calls for banks to avoid and tell about financial crimes. Mandates the protection of consumer financial information. Oversee cybersecurity and risk management in the banking industry. To that end, changes will need to be made to the compliance framework that will need to be adopted to deal with the following facets of quantum adoption [40]. Quantum computing applications can't compromise sensitive customer data – that's a data privacy concern for banks. With quantum research being global, secure quantum banking needs to be developed on an international basis to manage cross border security risks. If quantum powered attacks break into banking systems, legal questions

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as to who is responsible will be asked [41].

While it will remain up to the regulators to set clearer guidelines on quantum security; even when they do, banks will still be unsure about how to integrate quantum solutions in a compliant manner. However, there are financial, security and regulatory barriers against the adoption of quantum computing in US banking despite its potential. Banks have to contend with the high infrastructure costs, emerging quantum related cybersecurity threats, and evolving RA frameworks. Yet, financial institutes can mitigate these risks by preparing themselves ahead of such encryption resistance, by cooperating with quantum experts, and interacting with regulators [42].

THE ROAD AHEAD: PREPARING FOR A QUANTUM-POWERED BANKING INDUSTRY

With quantum computing on the rise, US banks need to act, ahead of the game, to get ready for integrating quantum into financial security and fraud prevention tasks. Despite its early status, the technology is something that financial institutions can't afford to wait for until quantum computing becomes normal. The focus of the banks should rather be to develop resilience against quantum threats, to build a quantum ready security framework and to invest in collaborative innovation [43].

One of the initial steps banks need to start transitioning is the adoption of Post Quantum Cryptography (PQC) to protect critical data from quantum available cyber-attacks. This involves: Banks will have to audit their existing cryptographic infrastructure to find weaknesses that quantum computing can infiltrate. Since banks have started testing and integrating PQC algorithms, the National Institute of Standards and Technology (NIST) is working towards standardizing PQC algorithms that they can begin applying [44].

The transition phase needs to be done in such a way that maximum security is achieved, by combining classical and quantum resistant encryption is advised by many experts and this is achieved by developing hybrid encryption models. US banks can continue to be ahead of 'Q Day,' the point when quantum computers become strong enough to demolish orthodox encryption if they move now [45].

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QUANTUM AI FOR FRAUD DETECTION: INVESTING

With quantum enhanced artificial intelligence (QAI), banks have a chance to revolutionize fraud detection with quantum computing. Investing in quantum machine learning (QML) enables banks to do the following. Quantum computing aids in process of real time transaction data faster – You can process large datasets in real time to determine suspicious activities before fraud happens [46]. Quantum AI can identify patterns of hidden fraud by analyzing complex relationships within financial data it is more difficult for criminals to pass undetected by fraud detection systems. Traditional fraud detection logs often mistake legitimate transactions as fraudulent. Accuracy can be improved by the quantum AI making your customer experience smoother [47].

Those that begin the experimentation of QML based fraud prevention today will be immediately in a position of competitive advantage in terms of security and efficiency. The field of quantum computing is very elite and per se is not something financial institutions yet have the expertise for. Partner with the quantum technology companies Team up with IBM, Google and emerging quantum startups to speed up research and development. Become close allies to regulatory agencies [48]. The Federal Reserve, SEC and the NIST, as these IndoAmerican banks work with them to understand security in quantum technology so compliance is never an issue. We need to train the cybersecurity and IT professionals – we will need a workforce equipped with the knowledge of how to use quantum AI and quantum cryptography, even in the long run [49].

By employing these strategic initiatives, banks can future proof their operations and make a smooth move to quantum powered security. Getting ready for a quantum powered banking industry is not simply about new technology, it is about the whole strategy of encryption upgrades, AI investments, workforce training, and acquiring open leverage of industry collaboration [50]. US banks that move now will instead be prepared to confront new financial threats, improve fraud prevention, and set the pace for secure banking in the future, before fully quantum computing is a reality.

CONCLUSION

The deployment of quantum computing in the US banking industry is a game changer for how the financial institutions guard against fraud and enhance the cybersecurity. With financial crimes

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becoming increasingly sophisticated, the prospect of quantum computing is helping banking Sector Company's stay ahead of the curve in battling these crimes. Quantum technology is capable of processing enormous amounts of data, which can unearth concealed fraud modulations at boggling paces, and it can boost the quality of a fraud discovery, real time transaction examining, and encryption security.

But reaching a banking system powered by a quantum computer is no small issue. This includes high costs in implementing quantum infrastructure, where new encryption standards are necessary, and finally, security threats from quantum powered cyber-attacks. Moreover, even for the US banks, regulatory environment for quantum technology in banking is changing rapidly and the banks must be agile and collaborate with respective regulators to comply with new standards.

Proactive investment in quantum technologies, post quantum cryptography adoption and quantum enhanced artificial intelligence algorithms to discover fraud prevention ways will pave the way ahead for the industry. In a secure quantum ready banking environment, collaboration between banks, technology innovators and the regulatory bodies will be required. However, by accepting these innovations, US banks can keep pace with new threat vectors and protect sensitive customer data and the integrity of the financial system from emerging threats. As quantum computing comes of age, the ability to change how financial security is defined means that it will become a vital tool for US banks to master. This will continue to be the case with the future evolution and responsible use of quantum computing technologies, which will represent the next period of more secure and resilient banking operations.

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